

Linear systems and elimination

The square-root upper bound for Gaussian partial-pivoting growth

Difficulty: challenging **Importance:** interesting to the community

Status: open; polynomial growth with high probability is known.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \leq j \leq n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let $G_n \in \mathbb{R}^{n \times n}$ have independent $N(0, 1)$ entries. Is the following precise square-root upper-bound conjecture true?

$$\text{For every } \eta > 0, \quad \lim_{n \rightarrow \infty} \Pr\{\rho_{\text{PP}}(G_n) > n^{1/2+\eta}\} = 0.$$

Here growth is measured over the exact-arithmetic Schur complements as defined above. This formulation asks only for the conjectured upper exponent; it does not add an unsupported matching lower-bound assertion or a limiting-distribution claim. It is also distinct from IE-04, which requires a uniform result after perturbing every deterministic center and prescribes an exponential tail.

References

Huang and Tikhomirov, *Average-case analysis of the Gaussian elimination with partial pivoting*, PTRF 189 (2024), 501–567, introduction, discussion of Edelman’s numerical evidence and main theorems. Trefethen and Schreiber, *Average-Case Stability of Gaussian Elimination*, SIAM J. Matrix Anal. Appl. 11 (1990), 335–360.

Status check

Searches for *Gaussian elimination* $n^{1/2}$ 2025 2026 and [site:arxiv.org](https://arxiv.org) *Gaussian growth factor* 2026 found polynomial upper bounds and the August worst-case results, but no proof at the square-root exponent. The distinction between exact and computed growth factors matters; the catalog statement fixes the former.